

Playwright Capstone Project Planner

Project Name: Enterprise Quality Engineering Automation Framework

Target Application: OrangeHRM Open Source Enterprise Demo

Selected Website: <https://opensource-demo.orangehrmlive.com/>

Document Version: 4.0 (Core Playwright & OrangeHRM Edition)

1. Application Context: Why OrangeHRM?

OrangeHRM is a world-class, open-source Human Resource Management (HRM) system utilized by enterprises globally. For this Capstone Project, OrangeHRM serves as the perfect target application because it naturally contains all the complex UI/UX components required to validate an engineer's Playwright capabilities.

- **Data-Heavy Grids:** The Employee (PIM) and Admin lists provide dynamic tables requiring advanced array iteration, sorting validation, and pagination handling.
- **Complex Form Elements:** The Recruitment and User Profile modules utilize customized non-select dropdowns, multi-step wizards, date pickers, and asynchronous file uploads.
- **SVG Graphics Rendering:** The main Dashboard relies on dynamic, coordinate-based SVG pie charts and timesheet graphs, fulfilling the "Project #2" low-level DOM manipulation requirement.
- **Network & API Architecture:** It utilizes a modern backend API, making it ideal for practicing `page.route()` backend mocking (e.g., intercepting JSON responses) and session cookie extraction for API-based logins.

2. Project Objectives

- **Scale:** Develop 120 automated test cases across 8 OrangeHRM modules (15 tests per module).
- **Architecture:** Use a reusable Page Object Model (POM) and data-driven testing.
- **Resiliency:** Write stable, parallel-ready tests with robust locators and no hard-coded waits.
- **Advanced Features:** Handle API mocking, file uploads, visual regression masking, and SVG extraction.
- **Reporting:** Generate advanced HTML results using Allure Report.

3. Technology & Automation Stack

Technology	Function in Framework
Playwright v1.4x	Core engine for interacting with OrangeHRM UI, extracting SVGs, and intercepting APIs.
JavaScript / Node.js	Core programming language and runtime environment.
Git & GitHub	Version control, branching strategies, and repository management.
Allure Report	Advanced reporting suite capturing DOM snapshots, videos, and analytics trends.
GitHub Copilot / GenAI	AI acceleration for refactoring brittle CSS into semantic getByRole locators.

4. Planned Modules / Services

- **Module 1: Authentication & Security**
Testing login workflows, session management, and bypassing the UI via API for faster test setup.
- **Module 2: Employee Data Grid / PIM** Verifying the employee table functionality, including sorting, pagination, and dynamic row actions.
- **Module 3: Recruitment Forms**
Automating candidate application forms, including complex dropdowns and resume file uploads.
- **Module 4: Dashboard Analytics**
Validating graphical dashboard widgets and extracting data from dynamic SVG charts.

- **Module 5: Leave Management**
Testing the leave application process, handling calendar date-picking, and browser dialog popups.
- **Module 6: "My Info" Profile**
Checking user profile updates, field limits, and profile picture upload/download functionalities.
- **Module 7: Time Tracking**
Performing visual regression testing on timesheets while masking dynamic clock and date values.
- **Module 8: Admin Panel**
Mocking backend API responses to test how the UI handles fake data injections and server errors.

5. Execution Timeline

Day	Planned Work
Day 1	Project setup, folder structure initialization, and configuration.
Day 2	Automate Authentication and Employee Data Grid modules.
Day 3	Automate Recruitment Forms and Dashboard Analytics modules.
Day 4	Automate Leave Management and "My Info" Profile modules.
Day 5	Automate Time Tracking (Visuals) and Admin Panel (API Mocking) modules.
Day 6	Code refactoring, test debugging, and pushing final code to GitHub.
Day 7	Generate Allure reports and prepare the final project documentation.

6. Framework Directory Structure

Playwright-OrangeHRM-Capstone/

```
├─ api/                # API bypass controllers (auth bypass, token
generators)
├─ data/                # Data-driven JSON arrays for form testing
├─ pages/              # Page Object Model classes
│   └─ components/     # Reusable widgets (Navbar, PaginationTable, SVGChart)
│   └─ AdminPage.js    # OrangeHRM Admin view mapping
│   └─ PIMPage.js      # OrangeHRM PIM view mapping
│   └─ RecruitmentPage.js
├─ tests/              # specs strictly grouped by module
│   └─ 01_auth.spec.js
│   └─ 02_pim.spec.js
│   └─ 03_recruitment.spec.js
│   └─ 04_dashboard.spec.js
│   └─ 05_leave.spec.js
│   └─ 06_myinfo.spec.js
│   └─ 07_time.spec.js
│   └─ 08_admin_mocking.spec.js
├─ utils/              # Custom hooks, soft-assertions, faker-generators
├─ screenshots/        # Baseline visual regression files
├─ playwright-report/  # Default HTML report output
├─ allure-results/     # Raw Allure data (processed later via 'allure
generate')
├─ playwright.config.js # Workers, retries, reporting, and parallel config
├─ .env                # Environment variables (Ignored in Git)
├─ .gitignore          ──
package.json
└─ README.md
```

Requirement Traceability Matrix (RTM)

Priority	Requirement / Feature Description	Target Module / Service	Test Cases	Test Case Examples
1 (High)	Secure Access & Session Management: Validate login workflows, session security, and enable API-based authentication bypass for framework speed.	Module 1: Authentication & Security	15	Valid/Invalid UI logins, forgot password workflows, session expiration, and API login bypassing.
2 (High)	Employee Data & Grid Functionality: Verify the extraction, sorting, and manipulation of tabular data arrays within the core HR database.	Module 2: Employee Data Grid (PIM)	15	Extracting tabular data, column sorting, multi-page pagination controls, and clicking dynamic row-level edit/delete actions.
3 (High)	User Profile Self-Service: Validate user profile updates, field boundary limits, and personal document upload/download streams.	Module 6: "My Info" Profile	15	Updating personal details, checking input field limits, uploading avatar images, and validating file downloads.
4 (Medium)	Candidate Application Processing: Automate complex form submissions, custom dropdowns, and asynchronous file uploads.	Module 3: Recruitment Forms	15	Navigating custom listboxes, uploading multi-file resumes (PDF/DOCX), and changing candidate application status.

5 (Medium)	Time-Off Workflows & Alerts: Validate leave request scheduling, complex calendar interactions, and native browser alert handling.	Module 5: Leave Management	15	Selecting cross-month date ranges, handling Accept/Reject native browser dialogs, and verifying leave balance deductions.
6 (Medium)	Timesheet Structural Layout: Perform pixel-perfect visual validation of timesheets while ignoring dynamic clock or timestamp data.	Module 7: Time Tracking (Visual)	15	Capturing baseline visual snapshots, masking live dates/clocks to prevent false test failures, and checking layout shifts.
7 (Low)	Graphical Data Rendering: Validate data accuracy and hover mechanics within dynamic SVG charts and complex dashboard widgets.	Module 4: Dashboard Analytics	15	Hovering over specific SVG pie chart nodes to extract data tooltips, and interacting with isolated Shadow DOM widgets.
8 (Low)	UI Resilience & Edge Cases: Validate how the UI behaves when backend APIs fail or return unexpected mock data.	Module 8: Admin Panel (API Mocking)	15	Mocking "No Users Found" API responses, simulating 500 Server Errors, and testing network timeout/loading spinner states.

7. Expected Outcomes

By the end of this project, the final repository will demonstrate:

- A fully functional, maintainable automation framework using the Page Object Model (POM).
- Automated test cases executing reliably in parallel without session interference.
- Practical mastery of complex automation challenges like tables, iframes, file uploads, and SVGs.
- Improved test execution speed using API mocking and programmatic login techniques.
- A comprehensive Allure dashboard displaying test trends, screenshots, and failure traces.